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Part 3 - Gradient Descent

Given

$$\hat{y} = m_1 x_1 + m_2 x_2 + b$$

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \qquad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$$

$$m = \begin{bmatrix} m_1 \\ m_2 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \qquad b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$n = 2$ (data points) $\qquad \alpha = 0.005$ (learning rate)

$$X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}$$

Derivation

$$\hat{y}_i = m_1 x_{i1} + m_2 x_{i2} + b$$

$$J = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2$$

$$\frac{\partial J}{\partial m_1} = \frac{1}{n} \sum_{i=1}^{n} 2(\hat{y}_i - y_i) \cdot \frac{\partial \hat{y}_i}{\partial m_1}$$

$$\frac{\partial \hat{y}_i}{\partial m_1} = x_{i1}$$

$$\Rightarrow \frac{\partial J}{\partial m_1} = \frac{2}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i) x_{i1} \quad , \quad \frac{\partial J}{\partial m_2} = \frac{2}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i) x_{i2}$$

$$\frac{\partial \hat{y}_i}{\partial b} = 1$$

$$\Rightarrow \frac{\partial J}{\partial b} = \frac{2}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i) \quad \Rightarrow \quad \frac{\partial J}{\partial m} = \frac{2}{n} X^T (\hat{y} - y)$$

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Since $n = 2$, $\frac{2}{n} = \frac{2}{2} = 1$

$$\Rightarrow m_{new} = m - \alpha \frac{\partial J}{\partial m}$$

$$b_{new} = b - \alpha \frac{\partial J}{\partial b}$$

Iteration I

$m_1 = -1$

$m_2 = 2$

$b = 1$

Step 1: $\hat{Y} = Xm + b$

$\hat{Y}_1 = (1)(-1) + (3)(2) + 1$

$= -1 + 6 + 1$

$= 6$

$\hat{Y}_2 = (4)(-1) + (10)(2) + 1$

$= -4 + 20 + 1$

$= 17.$

$$\hat{Y} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

Step 2: $e = \hat{Y} - Y$

$e_1 = 6 - 5 = \underline{1}$

$e_2 = 17 - 6 = \underline{11}$

$$J = \frac{1^2 + 11^2}{2}$$

$$= \frac{1 + 121}{2}$$

$$= \frac{122}{2}$$

$$= \underline{61}$$

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Step 3 : $\frac{\partial J}{\partial m} = X^T e$, $\frac{\partial J}{\partial b} = e_1 + e_2$

$$\frac{\partial J}{\partial m_1} = (1)(1) + (4)(11) +$$
$$= 1 + 44$$
$$= \underline{45}$$

$$\frac{\partial J}{\partial m_2} = (3)(1) + (10)(11)$$
$$= 3 + 110$$
$$= 113$$

$$\frac{\partial J}{\partial b} = 1 + 11$$
$$= \underline{12}$$

Step 4 : $m_1 = -1 - 0.005 \times 45$

$$= -1 - 0.225$$
$$= -1.225$$

$$m_2 = 2 - 0.005 \times 113$$
$$= 2 - 0.565$$
$$= 1.435$$

$$b = 1 - 0.005 \times 12$$
$$= 1 - 0.06$$
$$= 0.94$$

$$m = \begin{bmatrix} -1.2250 \\ 1.4350 \end{bmatrix} \quad b = \begin{bmatrix} 0.9400 \\ 0.9400 \end{bmatrix}$$

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Sylvie - Iteration part

Starting from $m = [-1.225, 1.435]^T$
and $b = [0.94, 0.94]^T$

the cost is $J = 10.11625$
after the update $m = [-1.3079, 1.2302]^T$
and $b = [0.9230, 0.9230]^T$

1. predictions

$$X_m = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.225 \\ 1.435 \end{bmatrix} = \begin{bmatrix} -1.225 + 4.305 \\ -4.9 + 14.35 \end{bmatrix} = \begin{bmatrix} 3.08 \\ 9.45 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 3.08 \\ 9.45 \end{bmatrix} + \begin{bmatrix} 0.94 \\ 0.94 \end{bmatrix} = \begin{bmatrix} 4.02 \\ 10.39 \end{bmatrix}$$

2. Error and cost

$$e = \begin{bmatrix} 4.02 - 5 \\ 10.39 - 6 \end{bmatrix} = \begin{bmatrix} -0.98 \\ 4.39 \end{bmatrix}$$

$$J = \frac{1}{2}\left((-0.98)^2 + 4.39^2\right) = \frac{0.9604 + 19.2721}{2} = \frac{20.2325}{2} = \underline{\underline{10.11625}}$$

3. Gradients

$$\frac{\partial J}{\partial m} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -0.98 \\ 4.39 \end{bmatrix} = \begin{bmatrix} -0.98 + 17.56 \\ -2.94 + 43.9 \end{bmatrix} = \begin{bmatrix} 16.58 \\ 40.96 \end{bmatrix}$$

$$\frac{\partial J}{\partial m} = -0.98 + 4.39 = 3.41$$

4. New parameters

$$m_2 = \begin{bmatrix} -1.225 \\ 1.435 \end{bmatrix} - 0.005 \begin{bmatrix} 16.58 \\ 40.96 \end{bmatrix} = \begin{bmatrix} -1.225 - 0.0829 \\ 1.435 - 0.2048 \end{bmatrix} = \begin{bmatrix} -1.3079 \\ 1.2302 \end{bmatrix}$$

$$b_2 = 0.94 - 0.005(3.411)$$
$$= 0.94 - 0.01705$$
$$= 0.92295$$
$$\approx \underline{\underline{0.9230}}$$

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Naomi Bamgbose

Starting from $m = [-1.3079, 1.2302]^T$

and $b = [0.9230, 0.9230]^T$

the cost is $J \approx 3.4221$,

After the update $m = [-1.3393, 1.1559]$

and $b = [0.9215, 0.9215]^T$

1. Predictions

$$X_m = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1.3079 \\ 1.2302 \end{bmatrix} = \begin{bmatrix} -1.3079 + 3.6906 \\ -5.2316 + 12.302 \end{bmatrix} = \begin{bmatrix} 2.3827 \\ 7.0704 \end{bmatrix}$$

$$\hat{y} = \begin{bmatrix} 2.3827 \\ 7.074 \end{bmatrix} + \begin{bmatrix} 0.9230 \\ 0.9230 \end{bmatrix} = \begin{bmatrix} 3.3057 \\ 7.9934 \end{bmatrix}$$

2. Error and Cost

$$e = \begin{bmatrix} 3.3057 - 5 \\ 7.9934 - 6 \end{bmatrix} = \begin{bmatrix} -1.6943 \\ 1.9934 \end{bmatrix}$$

$$J = \frac{(-1.6943)^2 + 1.9934^2}{2} = \frac{2.870652419 + 3.973643356}{2} = \frac{6.8}{2}$$

$$\approx 3.4221$$

3. Gradients

$$\frac{\partial J}{\partial m} = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} -1.6943 \\ 1.9934 \end{bmatrix} = \begin{bmatrix} -1.6943 + 7.9736 \\ -5.0829 + 19.934 \end{bmatrix} = \begin{bmatrix} 6.2793 \\ 14.851 \end{bmatrix}$$

$$\frac{\partial J}{\partial b} = 1.6943 + 1.9934 = 0.2991$$

4. New Parameters

$$m_3 = \begin{bmatrix} -1.3079 \\ 1.2302 \end{bmatrix} - 0.005 \begin{bmatrix} 6.2793 \\ 14.8511 \end{bmatrix} = \begin{bmatrix} -1.3079 - 0.0313965 \\ 1.2302 - 0.074256 \end{bmatrix}$$

$$\approx \begin{bmatrix} -1.3393 \\ 1.1559 \end{bmatrix}$$

$\hat{y} = [3.0499, 7.1233]$,

$e = [-1.9501, 1.1233]$

So $J = [3.802\,89001 + 1.261\,80289) / 2 = 2.5323$//

Result : the cost went from 3.4221 down to 2.5323, so the

update reduced the error.